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Temporal and spatial self supervised learning methods for electrocardiograms

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The limited availability of labeled ECG data restricts the application of supervised deep learning methods in ECG detection. Although existing self-supervised learning approaches have been applied to ECG analysis, they are predominantly image-based, which limits their effectiveness. To address these limitations and provide novel insights, we propose a Temporal-Spatial Self-Supervised Learning (TSSL) method specifically designed for ECG detection. TSSL leverages the intrinsic temporal and spatial characteristics of ECG signals to enhance feature representation. Temporally, ECG signals retain consistent identity information over time, enabling the model to generate stable representations for the same individual across different time points while isolating representations of different leads to preserve their unique features. Spatially, ECG signals from various leads capture the heart's activity from different perspectives, revealing both commonalities and distinct patterns. TSSL captures these correlations by maintaining consistency in the relationships between signals and their representations across different leads. Experimental results on the CPSC2018, Chapman, and PTB-XL databases demonstrate that TSSL introduces new capabilities by effectively utilizing temporal and spatial information, achieving superior performance compared to existing methods and approaching the performance of full-label training with only 10% of the labeled data. This highlights TSSL's ability to provide deeper insights and enhanced feature extraction beyond mere performance improvements. We make our code publicly available on https://github.com/cwp9731/temporal-spatial-self-supervised-learning.

Keywords Deep learning, Self-supervised learning, Electrocardiogram, Representation extraction

Cardiovascular diseases are one of the leading causes of death worldwide. According to statistics from the World Health Organization, cardiovascular diseases account for nearly one-third of all deaths globally each year[1]. As the incidence of cardiovascular diseases continues to rise, especially with a significant increase among younger populations, there is growing attention to their diagnosis and treatment[2]. Cardiovascular diseases are characterized by a long latency period, rapid onset, and severe sequelae, making timely and accurate diagnosis in the early stages crucial for effectively preventing heart attacks[3]. Electrocardiography (ECG) is a non-invasive method for detecting the electrical activity of the heart. It has been used in clinical cardiac disease diagnosis for over a century and has become an essential tool for physicians in diagnosing cardiovascular diseases[4].

In clinical practice, comprehensive ECG analysis and the ability to make rapid and accurate judgments typically rely on the deep professional knowledge and extensive clinical experience of cardiologists[5]. However, prolonged high-intensity work can lead to fatigue, increasing the risk of misdiagnosis and consuming significant medical resources. Furthermore, timely and accurate ECG diagnosis not only effectively prevents severe cardiovascular diseases but also enhances treatment outcomes. Unfortunately, manual diagnosis often struggles to meet this demand, leading to delays in medical attention when patients feel unwell. Consequently, as the importance of ECG diagnosis in medical practice continues to grow, the development of automated computer-aided diagnosis systems has become inevitable[6].

In recent years, deep learning methods, particularly supervised approaches, have made significant strides in the automatic detection of ECG signals.Supervised learning trains models using labeled data, enabling them to predict outputs based on input data. For example, Zhang et al.[7] combined convolutional neural networks with Transformers to develop a 12-lead ECG arrhythmia classification method, achieving advanced performance in ECG signal classification tasks. Similarly, Hannun et al.[8] utilized the ResNet architecture to

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create a 34-layer convolutional neural network (CNN) that demonstrated expert-level performance in single-lead ECG arrhythmia detection. These deep learning methods have shown high accuracy and low inference time in ECG signal detection, meeting the requirements for timeliness and accuracy in ECG diagnosis. However, the supervised training of these deep neural networks requires a large amount of labeled data, and the high cost and difficulty of obtaining ECG labels limit the widespread application of supervised learning methods in automatic ECG detection[9,10].

Self-supervised learning leverages the information contained within the data itself as supervisory signals to train deep neural networks, providing an effective solution for ECG detection scenarios where labeled data are difficult to obtain[11]. Self-supervised learning does not directly classify data; instead, it extracts features by training with the data's inherent information serving as labels. The process of encoding signals into features is commonly known as the pretraining process. These features can subsequently be applied to downstream tasks, such as ECG classification as discussed in this paper. If the pretrained model is directly transferred to the classification task, this is referred to as linear evaluation, which can reflect the performance of the extracted representations. If the pretrained model is further trained on the classification task, it is termed finetuning evaluation, which can assess the robustness and transferability of the learned representations. By utilizing the intrinsic information in unlabeled data, self-supervised learning can generate effective feature representations that require only a small amount of supervised data to achieve high performance in downstream tasks, thus reducing dependence on labeled data. Contrastive learning, a key method in self-supervised learning, involves aligning features generated from the same instance[12,13]. Typically, multiple views are created by applying different transformations (such as adding noise, rotation, cropping, flipping, etc.) to the same instance, treating them as positive pairs, while views of different instances are treated as negative pairs. The encoder is then trained to make the representations of positive pairs similar and those of negative pairs distinct. By using the intrinsic attributes of instances as supervisory signals instead of manual labels, contrastive learning effectively extracts essential features. However, commonly used data augmentation methods in the image domain (such as rotation, cropping, flipping, etc.) may disrupt the physiological information in ECG signals, thereby affecting the performance of downstream classification tasks[14]. Therefore, effectively utilizing the intrinsic attributes of ECG as supervisory signals to extract efficient and generalizable representations from ECG signals remains a significant challenge.

Building upon the principles of ECG signal generation, its unique medical characteristics, and insights from existing research, we identify the following temporal and spatial properties of ECG signals that our TSSL method leverages:

Temporal Properties (Temporal Invariance): In the field of biometric recognition, ECG signals have been shown to contain stable individual identity information that can be used for identification[15–17]. Figure 1 displays two segments of ECG signals collected from four individuals over different time periods in the ECG-ID database[18], with a maximum time span of up to 8 months. Despite physiological and psychological activities potentially causing local feature changes, the overall morphology of the ECG for the same individual remains highly similar. This indicates that the individual identity information in ECG signals demonstrates strong temporal stability, not significantly changing over time, thereby preserving the individual's identity characteristics. Therefore, ECG signals exhibit temporal invariance.

Spatial Properties (Spatial Correlation): As the heart beats, the electrical signals generated by myocardial cells combine to form a spatial vectorcardiogram (VCG), with the different ECG leads representing the projections of the VCG in various directions[19]. Medically, ECGs from different leads reflect the health status of different parts of the heart[20]. As shown in Fig. 1, ECGs from different leads exhibit similar overall trends, while each lead also possesses its own distinct characteristics. Therefore, ECGs demonstrate both similarity and distinctiveness among the different leads in the spatial dimension.

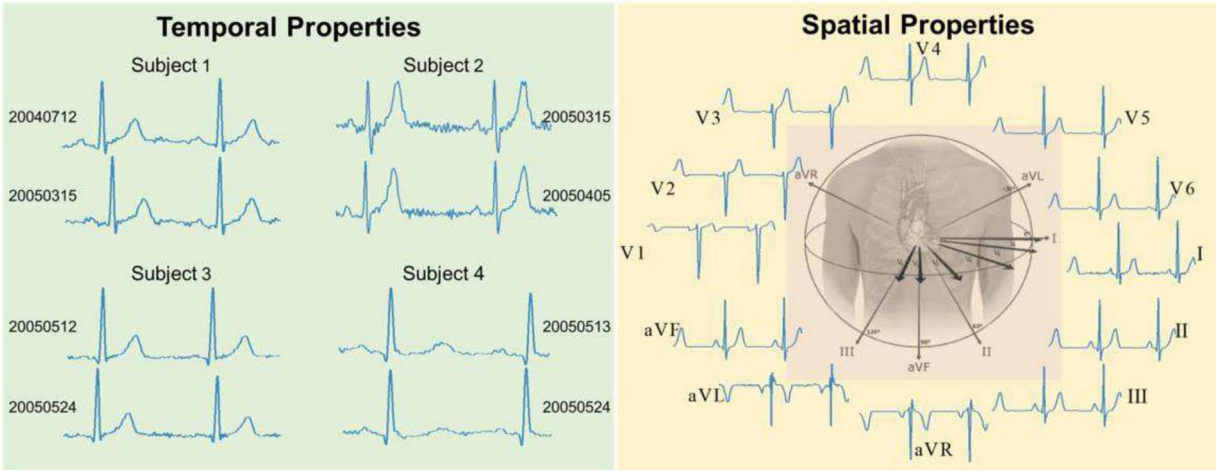


Fig. 1. Temporal-spatial properties of the ECG. Temporal invariance: ECG collected from the same individual at different times in the ECG ID database are highly similar. Spatial correlation: Different leads of the ECG represent observations of the heart from different spatial angles, and their changing trends are correlated.

In existing self-supervised learning methods for ECG signals, Kiyasseh et al.[21] and Diamant et al.[22] leveraged the temporal invariance of ECG signals, elevating the target of self-supervised learning from the traditional instance level to the individual level. They defined ECGs collected from the same individual at different times as positive pairs and ECGs from different individuals as negative pairs. Notably, Kiyasseh et al.[21] suggested that cardiac function is unlikely to undergo sudden changes within a short time frame (seconds), making temporal invariance valid over short periods but potentially invalid over longer durations (years). In their experiments, they used ECG data with short time spans. In contrast, Diamant et al.[22] argued that identity features in ECGs do not undergo significant changes due to time, pathology, or other factors, and therefore used long time span ECG data in their experiments. Experimental results showed that temporal invariance was effective across different time spans.

However, these methods overlook the information correlation between leads in multi-lead ECGs. Different leads have varying importance in the diagnosis of various conditions. For instance, among the 12 leads, lead II provides the clearest signal and performs exceptionally well in diagnosing ECG heart rate types, making it commonly used by doctors for quickly obtaining an initial impression of the ECG[23]. When diagnosing right bundle branch block (RBBB), the "m" or "RSR" complex waveform in lead V1 is a crucial feature, with lead V1 showing the best performance in RBBB diagnosis[24]. Although lead aVR is often overlooked in clinical practice, it holds significant value in diagnosing acute coronary syndrome and narrow complex tachycardia[25,26]. Additionally, lead I shows excellent results in detecting atrial fibrillation (AF), while posterior myocardial infarction (MI) typically presents in only four precordial leads[27,28]. Therefore, it is crucial to consider the information correlation between leads, as certain ECG patterns may appear simultaneously in different leads, indicating a lesion and its specific origin within the heart[29]. Neglecting this correlation when extracting latent features could impede the accurate diagnosis of ECGs.

To extract higher-quality and more insightful representations that enhance ECG classification, we introduce the Temporal-Spatial Self-Supervised Learning (TSSL) method. TSSL integrates two distinct self-supervised learning processes: Temporal Self-Supervision and Spatial Self-Supervision, each specifically designed to harness the temporal invariance and spatial correlations of ECG signals, respectively. Unlike previous studies[21,22] that primarily focus on temporal invariance at the instance or individual level, TSSL introduces a novel approach by applying lead isolation in temporal self-supervision. Specifically, ECGs collected from the same lead of the same individual at different times are treated as positive pairs, while those from the same lead of different individuals are treated as negative pairs. This isolation ensures that the unique characteristics of each lead are preserved without cross-lead interference. In spatial self-supervision, TSSL maintains the inherent correlations between different leads during representation extraction, ensuring that the spatial relationships and patterns across leads are consistently captured. By preserving both temporal and spatial integrity, TSSL not only improves feature extraction but also provides deeper insights into the ECG signal structure, facilitating more accurate and reliable diagnoses.

The main contributions of this work are summarized as follows:

Proposed an innovative temporal-spatial self-supervised learning (TSSL) method: Addressing the unique temporal and spatial characteristics of electrocardiogram (ECG) signals, this study designs two distinct learning processes: temporal self-supervision and spatial self-supervision. Temporal self-supervision isolates different leads of the same individual's ECGs collected at different times to preserve the uniqueness of each lead. Spatial self-supervision maintains the correlations between leads during feature extraction, ensuring effective integration of inter-lead information. This approach significantly enhances the effectiveness of feature extraction and improves the model's ability to capture complex patterns in ECG signals.

Introduced a lead isolation and correlation mechanism to enhance ECG diagnostic accuracy: In the temporal self-supervision phase, different leads' ECGs are isolated to prevent interference between lead representations, thereby preserving the unique information of each lead. In the spatial self-supervision phase, the correlations between leads are maintained to ensure that the extracted representations reflect the cooperative changes among leads. This dual mechanism effectively strengthens the model's understanding of the ECG's temporal and spatial attributes, thereby improving diagnostic accuracy and reliability.

Enhanced model generalization and practical applicability by eeducing dependence on extensive labeled data: Utilizing self-supervised learning methods, this study effectively leverages unlabeled ECG data to extract efficient and universally applicable feature representations. This approach not only significantly reduces the cost and difficulty associated with acquiring labeled data but also enhances the model's applicability and scalability in real-world medical settings. Compared to traditional supervised methods, TSSL maintains superior performance even with limited labeled data, demonstrating stronger practical value and potential for broader application.

Overcame limitations of existing self-supervised methods in ECG analysis, providing new research perspectives: Although previous studies have applied self-supervised learning to ECG analysis, they largely overlook the spatial correlations among multiple ECG leads. TSSL addresses this gap by simultaneously considering temporal invariance and spatial correlations, offering a more comprehensive feature extraction method that surpasses traditional approaches reliant solely on temporal attributes. This innovative design not only improves model performance but also provides new insights and methodologies for future research in multi-lead ECG signal self-supervised learning.

Related work

Self-supervised learning

Self-supervised learning has gained widespread attention because it can avoid the high costs associated with large-scale data labeling. It can be broadly categorized into generative methods and contrastive methods[30]. Generative methods include autoregressive models (AR)[31], flow-based models[32], autoencoder models (AE)[33], and hybrid models[34]. These models typically consist of an encoder and a decoder, and they can recover the

original data distribution without making assumptions about downstream tasks. In contrast, the purpose of contrastive learning is to distinguish whether two views represent the same instance. Typically, this only requires designing an encoder and using similarity measures to determine the proximity of two representations, thereby extracting high-quality representations[35].

Contrastive learning

In contrastive learning, different views of the same sample are defined as positive pairs, while views of different samples are defined as negative pairs. The loss function is constructed by maximizing the mutual information between different augmentations of the same sample in the latent space, which, in turn, trains the encoder. Thus, the key to contrastive learning lies in the definition of positive and negative pairs. Oord et al.[36] proposed Contrastive Predictive Coding (CPC), defining positive and negative pairs based on the predicted and actual outputs of the sample. Tian et al.[37], inspired by how humans perceive the world through multiple perspectives (such as eyes and ears), proposed Contrastive Multiview Coding (CMC), which defines positive and negative pairs based on the multiple views of a sample. He et al.[38] introduced the Momentum Contrast (MoCo) method, reframing previous contrastive learning approaches as a dictionary lookup problem by designing a dynamic dictionary and introducing a momentum encoder to update the encoder's parameters, achieving performance comparable to supervised training. Chen et al.[39] proposed a Simple Framework for Contrastive Learning (SimCLR), where different views are generated through data augmentation to define positive and negative pairs. They also introduced a projection head, projecting representations into the contrastive loss space for loss computation rather than using the representations directly for loss calculation. Grill et al.[40] proposed Bootstrap Your Own Latent (BYOL), which also generates different views through data augmentation but discards negative samples. Instead, the encoder's parameters are updated by learning from positive samples, demonstrating the feasibility of contrastive learning without negative samples.

Self-supervised learning applied to ECGs

Self-supervised learning methods applied to the automatic detection of ECGs are primarily based on contrastive learning. Mehari et al.[41] applied CPC[36], SimCLR[39], and BYOL[40] to self-supervised learning for ECG signals, exploring the impact of data augmentation methods, such as artificial transformations and ECG-specific physiological noise transformations, on the quality of representations. Gopal et al.[42] converted twelve-lead signals into a spatial vectorcardiogram (VCG)[43] using physiological excitation enhancement (3KG), and then transformed the VCG in three-dimensional space to generate multiple views for defining positive and negative pairs. Diamant et al.[22] proposed a patient contrastive learning method (PCLR), assuming that the individual characteristics of ECG signals remain constant over long periods and treating signals from the same patient at different times as positive pairs. Diamant et al.[21] proposed contrastive learning of cardiac signals across space, time, and patients (CLOCS), encouraging similar ECG signal representations from the same patient across different times and leads. Lan et al.[44] proposed an intra-inter subject self-supervised learning (ISSL) method. In addition to defining positive and negative pairs similarly to[21] and[22], they considered signal stationarity and more finely segmented the signals into multiple parts, using KPSS[45] to detect signal changes. Beyond contrastive learning methods, Zhang et al.[46] proposed a manipulated temporal-spatial reversal self-supervised learning method (MTSR), where an encoder is trained by using a predictor to forecast the type of signal transformation. Although the methods mentioned above have demonstrated the effectiveness of self-supervised learning in the ECG field, most of them have not been closely integrated with ECG medical diagnosis and have overlooked the correlation between leads in multi-lead ECGs. While CLOCS considered lead correlations, it assumed consistent representations across different leads, which could lead to the loss of lead-specific information during the self-supervised learning process, potentially hindering downstream tasks.

Methods

Overview

In this section, we provide a detailed description of our method. As shown in Fig. 2, our objective is to train an encoder through self-supervised learning on unlabeled ECG data, extract effective ECG representations, and evaluate their performance in downstream tasks. We represent the ECG dataset as $\mathscr{P} = \{(X_1, y_1), ..., (X_p, y_p)..., (X_P, y_P)\}$, where P is the total number of individuals included in the dataset. The ECG signal of individual p is represented as $X_p \in R^{C \times L}$, where C is the number of leads in the multi-lead ECG signal, L is the signal length, and y_p is the signal label.

Currently, a standard ECG is a 12-lead recording sampled at 500 Hz for 10 seconds, resulting in a signal length of 5000. To facilitate processing by deep learning models, we resample the signal length to 4096, thus setting $C = 12$, $L = 4096$ .Since it is challenging to obtain two ECG recordings from the same individual at different times, we achieve temporal self-supervised learning by splitting a continuously recorded ECG signal into two segments (this is reasonable, as two ECG signals from the same individual are more similar when recorded with a shorter time interval). At this point, the ECG signal X_p of individual p can be divided into $2 \times C$ segments x , with the i-th segment of the c-th lead is represented as $x_p^{c,i} \in R^{L/2}$.

In the self-supervised learning stage, we train an encoder f_θ to extract the latent representation $h_p^{c,i} \in R^E$ (Eq. 1) of each signal segment $x_p^{h,i}$, where E is the dimensionality of the extracted representation. To better optimize the learning process of the model, following the suggestions of Chen et al.[39] , we introduced a nonlinear projection headg_ω. The projection head is a small neural network module, typically consisting of one or two fully connected layers, which maps the representations $h_p^{c,i}$ generated by the encoder to a new space. In other words, instead of performing self-supervised learning directly on the representations, we use the nonlinear projection head g_ω to map the representations to a self-supervised loss space $z_p^{c,i} \in R^E$ (Eq. 2), where the self-supervised learning is performed to enhance the quality and discriminative power of the representations.

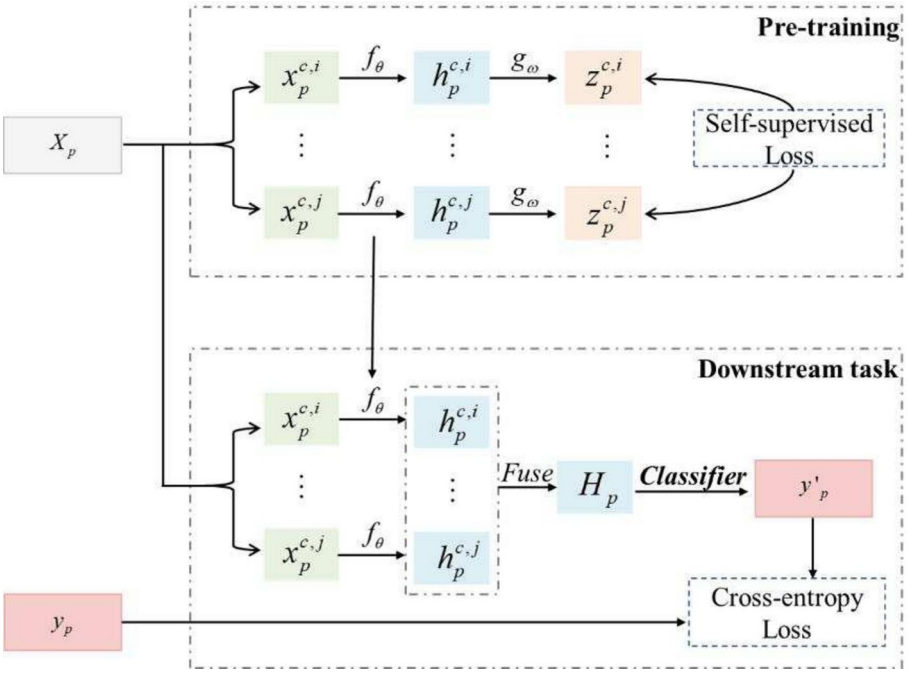


Fig. 2. Overall flow of the proposed method. Divided into two phases: pre-training and downstream tasks. Pre-training phase, different lead data $x_p^{h,i}$ are input into encoder f_θ and then passed through a projection head g_ω to project feature $h_p^{c,i}$ into the self-supervised loss space $z_p^{c,i}$ to calculate the loss. Downstream task phase, the encoder f_θ is transferred to this phase and then classified through a classifier.

$$h_p^{c,i} = f_\theta(x_p^{c,i}) \tag{1}$$

$$z_p^{c,i} = g_\omega(h_p^{c,i}) = \omega_2 \sigma_g(\omega_1 h_p^{c,i}) \tag{2}$$

where $\sigma_g(\cdot)$ is the ReLU activation function, defined as:

$$\sigma_g = \max(0, x) \tag{3}$$

The main objective of this paper is to design a high-performance self-supervised learning method. Therefore, we did not focus extensively on designing the encoder structure but instead directly employed several classic deep learning architectures to build the encoder. To better demonstrate the performance of the proposed self-supervised learning method, we selected four classic CNN architectures as encoders: LeNet[47], AlexNet[48], VGG[49], and ResNet[50]. The architecture of each model is illustrated in Fig. 3. In the figure, "Conv (Size)-Num" indicates the size and number of convolutional kernels in that layer. The self-supervised learning of encoder and projection head, as shown in Fig. 4, is discussed in Sections 3.2 and 3.3.

After completing the self-supervised learning, we transfer only the trained encoder f_θ to the downstream tasks, which, in this study, are all ECG signal classification tasks. The output of the multi-lead ECG signal X_p after passing through f_θ is the representation $\left\{h_p^{1,1} h_p^{1,2}, \ldots, h_p^{C,1} h_p^{C,2}\right\}$ of $2 \times C$ segments. First, the representations of all leads and segments are concatenated to obtain the complete representation H_p of X_p (Eq. 4), which is then input into a linear classifier to obtain the classification result $y\prime_p$ (Eq. 5).

$$H_p = Fuse(\{h_p^{1,1}, h_p^{1,2}, \ldots, h_p^{c,1}, h_p^{c,2}, \ldots, h_p^{C,1}, h_p^{C,2}\}) \tag{4}$$

$$y\prime_p = \sigma_l(q_{W,b}(H_p)) = \sigma_l(W H_p + b) \tag{5}$$

Here, W and b represent the weights and bias of the classifier, respectively, while $\sigma_l(\cdot)$ is the activation function of the classifier. In typical ECG classification tasks, the task types include single-label multi-class classification and multi-label multi-class classification. For single-label multi-class classification tasks, the activation function used is Softmax (Eq. 6), which outputs the probability distribution for each class.

$$\sigma_l(x) = e^{x_i} / \sum_i e^{x_i} \tag{6}$$

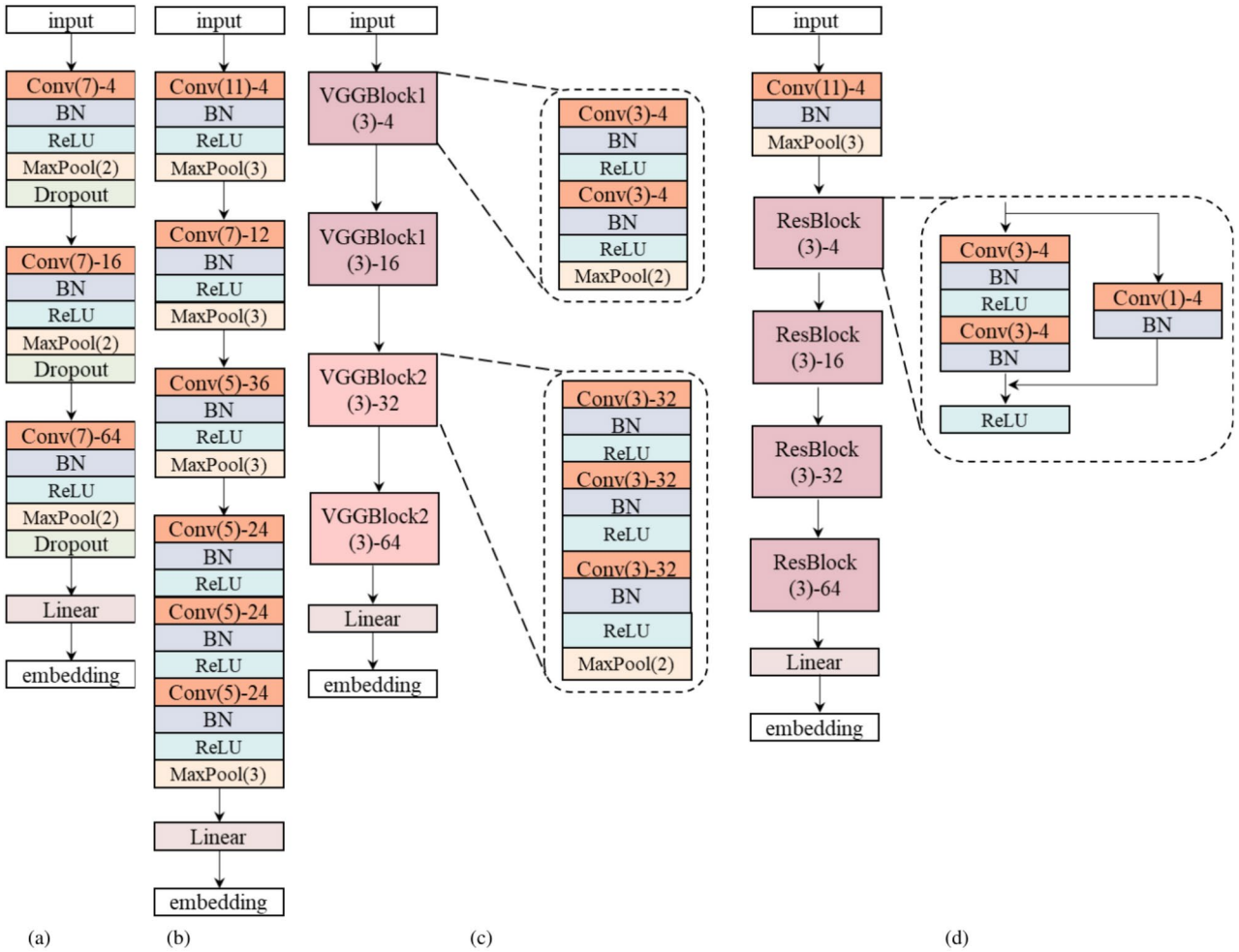


Fig. 3. Encoder architecture.

For multi-label classification tasks, each class can be treated as a binary classification task. Therefore, the Sigmoid activation function (Eq. 7) is used to independently calculate the probability for each class.

$$\sigma_l(x) = 1/(1 + e^{-x}) \tag{7}$$

By employing the appropriate activation function, the classifier can produce predictions according to the task requirements, thereby achieving optimal performance across different types of ECG classification tasks. Finally, the classification model is trained using the cross-entropy loss function L_{ce} (Eq. 8).

$$L_{ce} = -\frac{1}{N}\sum_{p=1}^{N}[y_p \log y\prime_p + (1 - y_p)\log(1 - y\prime_p)] \tag{8}$$

Here, N represents the total number of samples, y_p denotes the actual label of individual p, and $y\prime_p$ represents the model's predicted classification. By minimizing the cross-entropy loss function, the classification model can be effectively optimized to achieve higher accuracy in ECG classification tasks.

Temporal self-supervised learning

The ECGs collected from individual p_1 at different times are $(x_{p_1}^{v1}, \mathrm{x}_{p_1}^{v2})$, which, after being processed by the encoder f_θ and the nonlinear projection head g_ω , yield the representations $(x_{p_1}^{v1}, \mathrm{x}_{p_1}^{v2})$ and their projections in the self-supervised loss space$(z_{p_1}^{v1}, z_{p_1}^{v2})$.According to temporal invariance, the ECGs $(x_{p_1}^{v1}, \mathrm{x}_{p_1}^{v2})$ of the same individual at different times are usually defined as positive pairs, while the ECGs $(x_{p_1}^{v*}, \mathrm{x}_{p2}^{v*})$ of different individuals are defined as negative pairs. However, this approach can often disrupt the diversity of information carried by different leads, which is detrimental to the diagnosis of multi-lead ECGs. Therefore, we applied isolation between leads, defining the ECGs $(x_{p_1}^{c,v1}, \mathrm{x}_{p_1}^{c,v2})$ collected from the same lead of the same individual

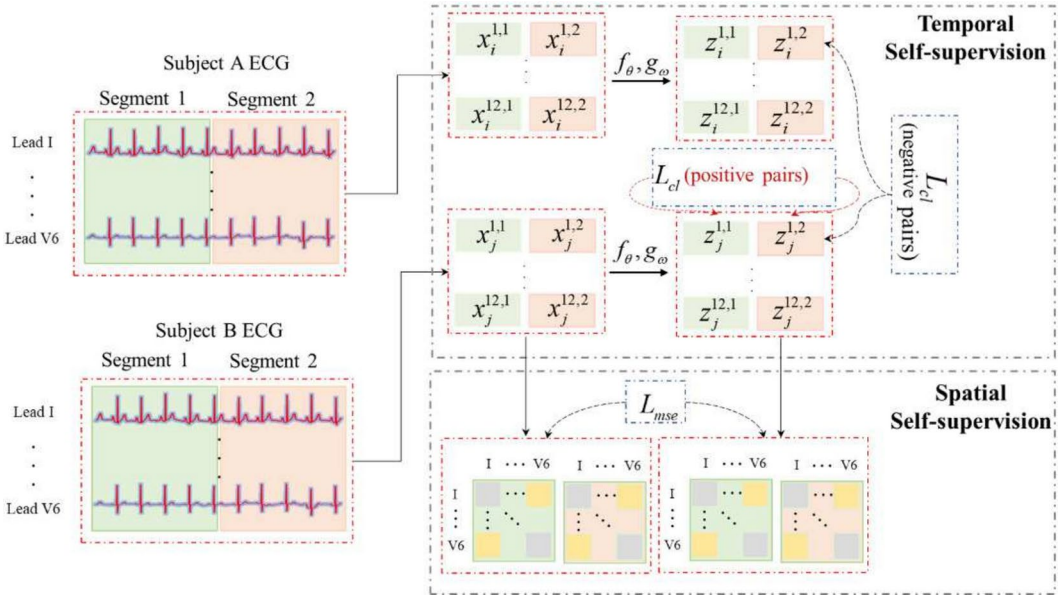


Fig. 4. Overview of self-supervised learning. Divided into temporal self-supervision and spatial self-supervision. Temporal self-supervision: The same lead from the same individual forms positive pairs, different leads from different individuals form negative pairs, and leads are isolated from each other. Spatial self-supervision: Calculate the correlation between different leads of the input signals and the correlation between different lead representations separately, using MSE loss for fitting.

at different times as positive pairs and the ECGs $(x_{p_1}^{c,v*}, \mathrm{x}_{p_2}^{c,v*})$ from the same lead of different individuals as negative pairs. The calculations for each lead are conducted independently without influencing each other.

After defining the positive and negative pairs, we use the normalized temperature-scaled cross-entropy loss (NT-Xent)[51,52] to define the temporal self-supervised learning loss L_{cl} (Eq. 9). This loss function optimizes the training process by maximizing the similarity of positive pair representations and minimizing the similarity of negative pair representations. By optimizing this loss function, the temporal invariance of ECGs can be better leveraged to extract effective representations.

$$L_{cl} = -\sum_{c=1}^{C}\sum_{p=1}^{B} \log \frac{\exp(s(z_p^{c,v1}, z_p^{c,v2})/\tau)}{\sum_{k=1}^{2B} \mathbb{I}_{[k\neq p]} \exp(s(z_p^{c,v*}, z_k^{c,v*})/\tau)} \tag{9}$$

Here, $\mathbb{I}_{[k\neq p]}$ is an indicator function that equals 1 (if and only if $k \neq p$ is true), B is the batch size, τ represents the temperature coefficient, typically set to 0.1 and $s(u, v)$ (Eq. 10) is the cosine similarity between vector u and vector v, defined as:

$$s(u, v) = \frac{u \cdot v}{|u||v|} \frac{1}{\tau} \tag{10}$$

Spatial self-supervised learning

Different ECG leads represent observations of the heart from various angles, and the diagnosis of cardiovascular diseases typically requires the integration of information from multiple leads to provide a more accurate diagnosis than using a single lead alone. Therefore, we established the goal of spatial self-supervised learning: to maintain the correlation of information between the input signals and their latent features across different leads during the self-supervised learning process. For an individual p whose signals and representation projections from different leads are $(x_p^{c_1,i}, x_p^{c_2,i})$ and $(z_p^{c_1,i}, z_p^{c_2,i})$, respectively, we use cosine similarity (Eq. 10) to measure the correlation of information between the different leads. To preserve this correlation during self-supervised learning, we minimize the difference between the cosine similarity $s(x_p^{c_1,i}, x_p^{c_2,i})$ of the original signals and the cosine similarity $(z_p^{c_1,i}, z_p^{c_2,i})$ of the representation projections using mean squared error (MSE). This approach ensures that the information correlation between leads is maintained throughout the learning process. The spatial self-supervised loss is defined in (Eq. 11).

$$L_{mse} = \frac{1}{2 * B * C * C} \sum_{p=1}^{B}\sum_{i=1}^{2}\sum_{c_1=1}^{C}\sum_{c_2=1}^{C} \mathbb{I}_{[c_1\neq c_2]} \left(s(x_p^{c_1,i}, x_p^{c_2,i}) - s(z_p^{c_1,i}, z_p^{c_2,i})\right)^2 \tag{11}$$

By minimizing this loss function, we can maintain the consistency of information correlation between leads, thereby preserving and leveraging the inter-lead correlation during the self-supervised learning process. Finally, by combining temporal and spatial self-supervision, we define the temporal-spatial self-supervised learning loss L_{sl} (Eq. 12). The self-supervised learning process is illustrated in Algorithm 1.

$$L_{sl} = L_{cl} + L_{mse} \tag{12}$$

Input: unlabeled dataset $X = \{X_1, ..., X_p, ..., X_P\}$, batch size$B$, encoder f_θ, project head g_ω
Output: f_θ, g_ω

Initialize f_θ, g_ω
for sampled minibatch$\{X_p\}_{p=1}^{B}$ **do**
  for $p = 1, 2, ..., B$ **do**
    Divide signal X_p to $\{x_p^{1,1}, x_p^{1,2}, ..., x_p^{C,1}, x_p^{C,2}\}$
    for $c = 1, 2, ..., C$ **do**
      for $i = 1, 2$ **do**
        $h_p^{c,i} = f_\theta(x_p^{c,i})$
        $z_p^{c,i} = g_\omega(h_p^{c,i})$
      end for
    end for
  end for
  for $p = 1, 2, ..., B$ **do**
    for $i = 1, 2$ **do**
      for $c_1 = 1, 2, ..., C$ **do**
        for $c_2 = 1, 2, ..., C$ **do**
          Calculate $s(x_p^{c_1,i}, x_p^{c_2,i})$ using Eq. 10
          Calculate $s(z_p^{c_1,i}, z_p^{c_2,i})$ using Eq. 10
        end for
      end for
    end for
  end for
  Get L_{cl}, $L_{mse}$by Eq. 9, Eq. 11
  Get $L_{sl}$by Eq. 12
  $\theta \leftarrow \arg\min_\theta L_{sl}, \omega \leftarrow \arg\min_\omega L_{sl}$
end for
return Outputs

Algorithm 1. Self-supervised training procedure of TSSL

Experiments and results

Datasets

We validated the performance of the TSSL model using three publicly available ECG datasets. Each dataset was divided into training, validation, and test sets in a 6:2:2 ratio. The size of each dataset is shown in Table 1, with specific details provided below:

Chapman[53]: The Chapman database, sponsored by Chapman University and Shaoxing People's Hospital (Zhejiang University Shaoxing Hospital), contains 10,646 12-lead ECG recordings with a sampling frequency of 500 Hz and a duration of 10 seconds. This dataset includes 11 common heart rhythm types, which we merged into 4 categories according to the recommendations of Lan et al.[44] and Kiyasseh et al.[21].

CPSC2018[54]: The China Physiological Signal Challenge 2018 (CPSC2018) database contains 6,877 12-lead ECG recordings, with a sampling frequency of 500 Hz and a recording length ranging from 6 to 60 seconds. We truncated these recordings to a uniform length of 10 seconds and removed those that were less than 10 seconds long. This dataset contains 9 types of heart rhythms.

PTB-XL[55]: The Physikalisch Technische Bundesanstalt XL (PTB-XL) database comprises 21,837 12-lead ECG recordings from 18,885 patients, with a sampling frequency of 500 Hz and a duration of 10 seconds, similar to the Chapman database. The dataset contains five diagnostic labels.

Dataset	Train	Val	Test	Label type	Categories
Chapman	6379	2121	2128	Single-label	4
CPSC2018	5660	1849	1855	Multi-label	9
PTB-XL	12900	4260	4270	Multi-label	5

Table 1. Details of dataset split.

All 10-second ECG data obtained from the three databases were only preprocessed through normalization before being fed into the model for training and testing.

Experiment setup

We train the encoder through self-supervised learning and transfer it to downstream tasks to extract representations of the input signal. The performance in these downstream tasks is then evaluated to assess the effectiveness of the representation extraction. In ECG application scenarios, downstream tasks are typically arrhythmia classification tasks. We employ the following two methods to perform downstream tasks:

Linear evaluation (Linear): In this approach, the encoder transferred to the downstream task is frozen, meaning its parameters are no longer updated, and only a linear classifier is trained. Following the recommendations of[21,44], the encoder and classifier are trained on the same dataset, with the encoder's performance in the downstream task reflecting the effectiveness of its representation extraction.

Transferability of representations (Fine-tuning): The encoder transferred to the downstream task is not frozen; instead, both the encoder and classifier are trained together. This approach allows for evaluating the robustness and transferability of the learned representations. In this scenario, the encoder is initially trained on one dataset and then fine-tuned in a supervised manner on other datasets.

During model training, the batch size is set to 128, and the Adam optimizer[56] is used to optimize parameters with a learning rate of 1e-3. The maximum number of epochs is set to 400, and training is terminated if the validation loss does not improve within 15 epochs, with the best validation loss model saved. The experiments in this paper were conducted on a machine with an NVIDIA GeForce 1050Ti GPU using the PyTorch[57] framework. Since trained deep learning models exhibit some degree of randomness, each model was trained 5 times, and the average ROC and standard deviation are reported as the final results.

Impact of data augmentation

Most existing self-supervised learning methods for ECG signals primarily use the SimCLR[39] framework. SimCLR's main approach involves applying different data augmentation techniques to the same instance, generating different views to define positive and negative pairs. However, this paper redefines positive and negative pairs from the instance level to the individual level, where signals from the same individual at different times are considered different views. Therefore, in our method, data augmentation is not a necessary step. To investigate the impact of data augmentation on the performance of our method, we compared the effects of various data augmentation methods on SimCLR and the proposed TSSL.

We used 7 data augmentation methods, including four artificial transformations[41]: Random Resized Crop (RRC), Time Out (TO), Gaussian Noise (GN), Gaussian Blur (GB); and three physiological noise transformations: Baseline Wander (BW), Electrode Motion (EM), and Muscle Artifact (MA). Among these, the physiological noise used in the physiological noise transformations was obtained from the MIT-BIH Noise Stress Test Database[58]. The effects of each transformation are illustrated in Fig. 5.

In this section, we use LeNet, as shown in Fig. 3, as the encoder, and perform downstream tasks on three datasets using two methods, resulting in nine downstream tasks: three linear tasks and six fine-tuning tasks. The experimental results are shown in Fig. 6, where "Linear: Chapman" refers to the linear evaluation performed on

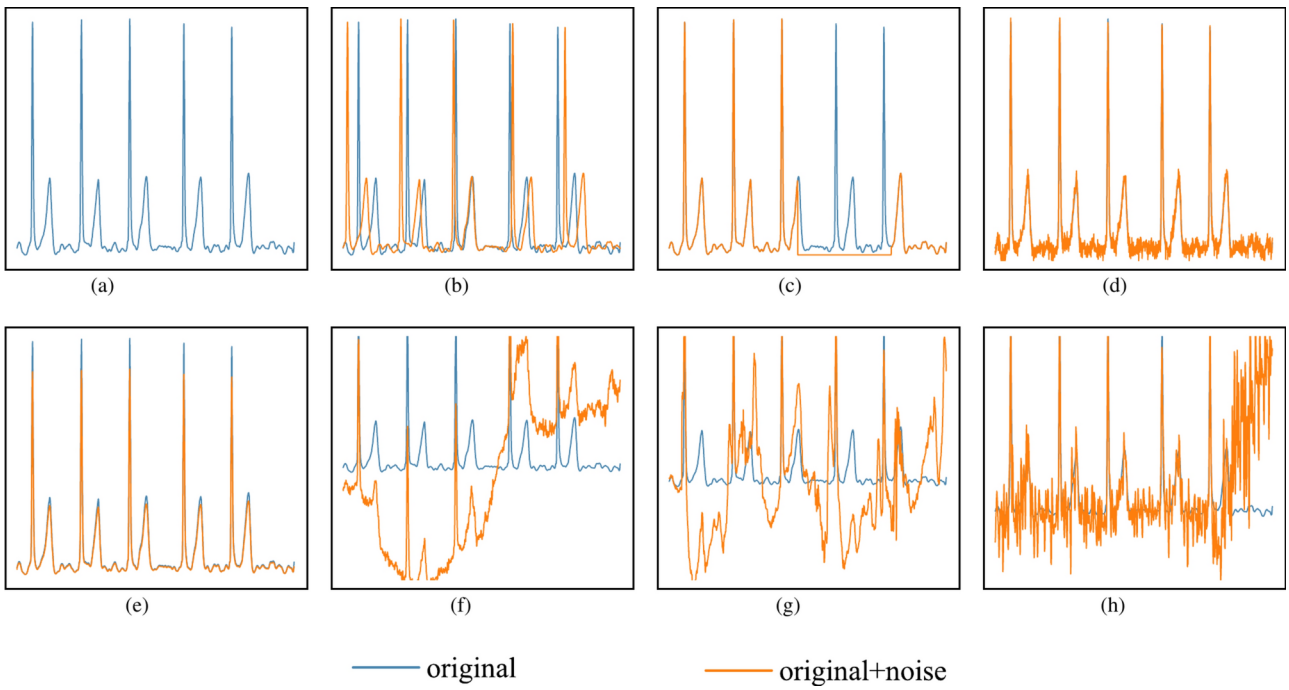


Fig. 5. ECG transformations. Comparison between the effects of different transformation methods on signals.

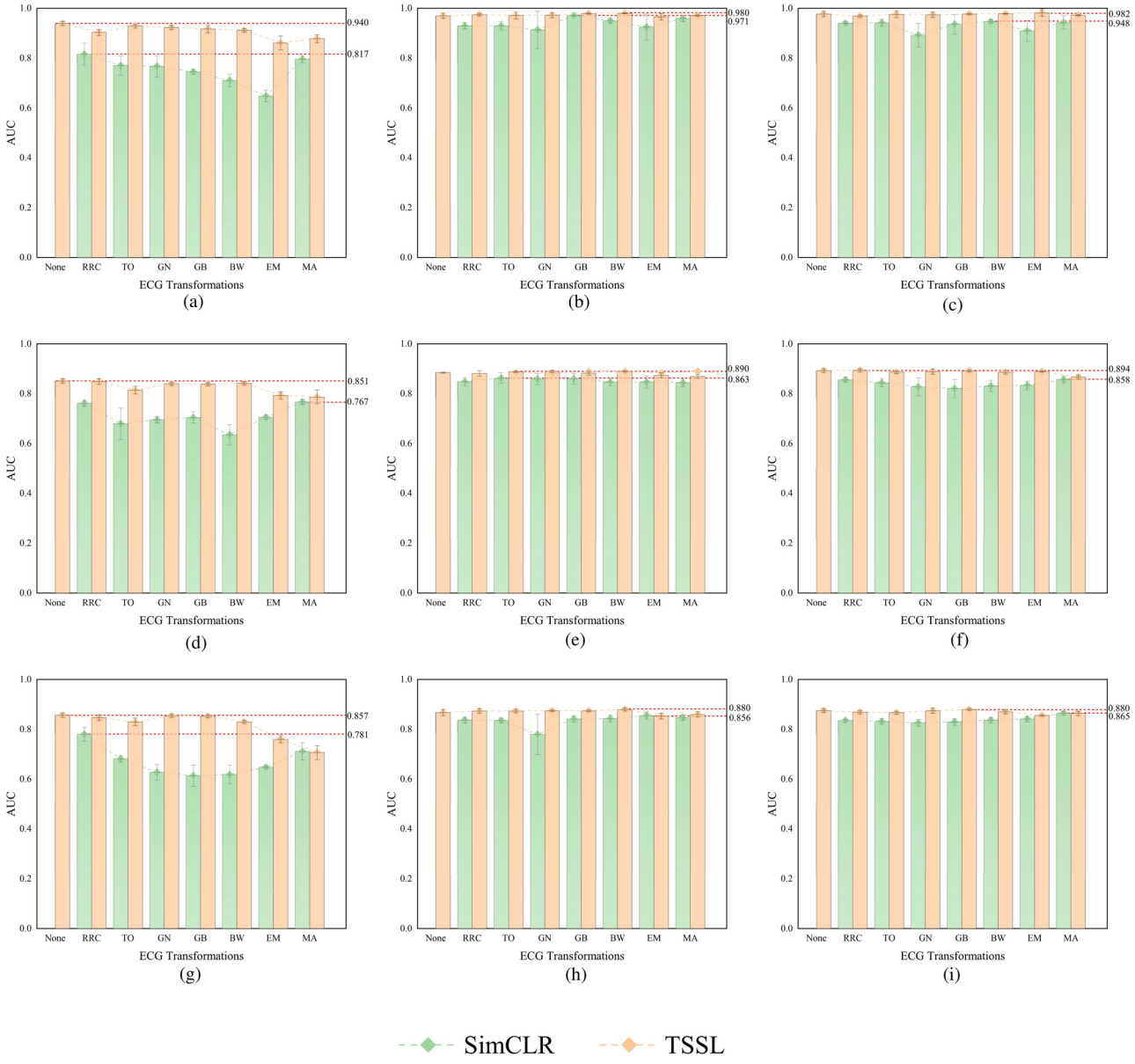


Fig. 6. Results of different data augmentation methods on SimCLR and TSSL. "Linear: Chapman" means that the encoder is trained on the Chapman dataset, then the encoder parameters are frozen and transferred to downstream tasks to train the classifier on the Chapman dataset. "Fine: CPSC2018→Chapman" means that the encoder is trained on CPSC2018, then transferred to downstream tasks to train both the encoder and classifier on the Chapman dataset.

the Chapman dataset. In this case, the encoder is first trained and then frozen while training the classifier. "Fine: CPSC2018→Chapman" indicates that the encoder is first trained on CPSC2018 and then transferred to the Chapman dataset for fine-tuning. Since data augmentation is not necessary for TSSL, additional results without data augmentation (None) are also presented for TSSL.

As shown in Fig. 6, no single data augmentation method demonstrates absolute superiority across all tasks. In linear tasks, SimCLR performs best with RRC on the Chapman and PTB-XL datasets, while it excels with MA on the CPSC2018 dataset. In contrast, TSSL without data augmentation performs optimally in linear tasks, suggesting that data augmentation may disrupt the extraction of certain diagnostic features. However, in fine-tuning tasks, although SimCLR with RRC excels in linear tasks, it does not exhibit the same advantage here. In contrast, TSSL with data augmentation shows some improvement over TSSL without data augmentation, indicating that noise can enhance the model's ability to generalize to different data distributions and improve the robustness of the representations.

From the above analysis, it is evident that SimCLR's performance is closely tied to the choice of data augmentation method, with significant variation in results across different methods. Designing a universal data augmentation strategy that works well in all scenarios is challenging. In contrast, TSSL without data

Data augmentation	Rank	
	SimCLR	TSSL
None	–	**6.08(1)**
RRC	4.50(2)	4.17(6)
TO	4.33(3)	4.71(5)
GN	3.00(6)	4.79(4)
GB	3.75(5)	5.42(3)
BW	2.75(7)	5.50(2)
EM	4.25(4)	3.25(7)
MA	**5.42(1)**	2.08(8)

Table 2. Friedman test of different data augmentation methods. Significant values are in bold.

Dataset	Chapman	CPSC2018	PTB-XL
Supervised	0.958±0.012	0.856±0.014	0.844±0.013
Random Init	0.755±0.019	0.729±0.010	0.688±0.003
SimCLR	0.797±0.015	0.767±0.010	0.712±0.035
BYOL	0.821±0.011	0.759±0.011	0.737±0.019
CLOCS	0.936±0.016	0.827±0.020	0.802±0.026
MTSR	0.773±0.038	0.763±0.046	0.753±0.010
3KG	0.904±0.015	0.707±0.033	0.715±0.058
TSL	0.929±0.018	0.839±0.013	0.842±0.010
SSL	0.929±0.016	0.753±0.009	0.735±0.038
TSSL	0.940±0.008	0.851±0.008	0.857±0.008

Table 3. Linear evaluation results using the LeNet encoder.

augmentation (None) demonstrates excellent performance across all tasks. Although using data augmentation in fine-tuning tasks offers a slight improvement over not using it, the gains are limited, and no single fixed data augmentation method consistently outperforms the no augmentation approach across all tasks.

The robustness of different data augmentation methods across various tasks is relatively poor, making it challenging to intuitively rank the performance of these methods. In the field of intelligent optimization algorithms[59], the Friedman test is commonly used to evaluate the overall performance of optimization algorithms across multiple tasks. Therefore, we used the Friedman test to evaluate the overall performance of different data augmentation methods across 9 tasks. The evaluation results are shown in Table 2. Since a higher ROC value indicates better performance, equivalent to maximizing the optimization task, a higher mean rank indicates better method performance. As shown in the table, the MA transformation yielded the best performance for SimCLR, so subsequent experiments with SimCLR will use the MA transformation. For TSSL, the best performance is achieved without using data augmentation; therefore, the TSSL method does not utilize data augmentation.

Comparison of methods

In this section, we use the four encoders from Fig. 3 to compare TSSL with state-of-the-art self-supervised learning methods for ECG, including SimCLR[39], BYOL[40], CLOCS[21], MTSR[46], and 3KG[42]. Additionally, to better highlight the performance of self-supervised learning, we also compare against encoders trained using supervised learning (Supervised) and randomly initialized encoders (Random Init). We also present the results of using temporal self-supervision (TSL) and spatial self-supervision (SSL) alone, conducting an ablation analysis to verify the effectiveness of the TSSL strategy. For a more thorough analysis, we ranked the models by complexity from simple to complex as LeNet, AlexNet, VGG, and ResNet, and ranked the datasets by classification difficulty from simple to complex as Chapman, CPSC2018, and PTB-XL.

Linear evaluation

Tables 3, 4, 5 and 6 present the linear evaluation results for different methods using the 4 encoders, with the bolded values representing the best results (supervised methods are included only for reference and are not part of the comparison). First, the tables show that TSSL outperforms other algorithms regardless of whether a simple or complex network structure is used as the encoder, indicating TSSL's strong robustness across different network structures and its consistent ability to achieve excellent performance. Secondly, on the simpler Chapman dataset, TSSL's advantage over other algorithms is not significant. For instance, in Table 5, TSSL shows only a 1.7% improvement compared to 3KG and CLOCS. However, on the more complex PTB-XL dataset, TSSL shows improvements of 16.5% and 15.7% over 3KG and CLOCS, respectively, further demonstrating TSSL's superior performance in feature extraction.

Dataset	Chapman	CPSC2018	PTB-XL
Supervised	0.976±0.004	0.868±0.011	0.868±0.011
Random Init	0.755±0.019	0.744±0.010	0.674±0.027
SimCLR	0.741±0.008	0.722±0.009	0.705±0.007
BYOL	0.766±0.021	0.733±0.011	0.746±0.022
CLOCS	0.947±0.006	0.820±0.020	0.805±0.015
MTSR	0.732±0.025	0.719±0.013	0.698±0.044
3KG	0.911±0.012	0.842±0.010	0.788±0.030
TSL	0.941±0.005	0.852±0.023	0.854±0.003
SSL	0.941±0.004	0.767±0.012	0.754±0.030
TSSL	0.956±0.010	0.882±0.004	0.876±0.004

Table 4. Linear evaluation results using the AlexNet encoder.

Dataset	Chapman	CPSC2018	PTB-XL
Supervised	0.988±0.002	0.851±0.010	0.862±0.013
Random Init	0.763±0.045	0.734±0.016	0.682±0.017
SimCLR	0.777±0.019	0.741±0.002	0.723±0.066
BYOL	0.769±0.016	0.728±0.018	0.684±0.030
CLOCS	0.967±0.008	0.763±0.017	0.758±0.014
MTSR	0.616±0.077	0.664±0.022	0.647±0.036
3KG	0.968±0.007	0.830±0.022	0.747±0.021
TSL	0.973±0.003	0.863±0.009	0.851±0.004
SSL	0.974±0.000	0.805±0.010	0.784±0.087
TSSL	0.984±0.003	0.875±0.003	0.877±0.007

Table 5. Linear evaluation results using the VGG encoder.

Dataset	Chapman	CPSC2018	PTB-XL
Supervised	0.986±0.001	0.818±0.011	0.858±0.009
Random Init	0.732±0.032	0.707±0.013	0.646±0.017
SimCLR	0.699±0.023	0.578±0.020	0.648±0.008
BYOL	0.710±0.014	0.658±0.029	0.644±0.018
CLOCS	0.958±0.009	0.761±0.022	0.784±0.017
MTSR	0.628±0.054	0.634±0.046	0.583±0.060
3KG	0.926±0.010	0.836±0.009	0.751±0.008
TSL	0.958±0.003	0.865±0.010	0.852±0.004
SSL	0.863±0.012	0.766±0.012	0.749±0.028
TSSL	0.970±0.007	0.878±0.007	0.872±0.009

Table 6. Linear evaluation results using the ResNet encoder.

Additionally, as seen in Tables 3, 4, 5 and 6, TSSL outperforms supervised training on the CPSC2018 and PTB-XL datasets, which is a surprising result. This is because self-supervised learning lacks direct label guidance and a clear task objective, whereas supervised learning typically delivers better results when labeled data is available. To investigate the reason behind this phenomenon, Fig. 7 shows the changes in training loss, validation loss, and test loss during the classifier training process after using the VGG encoder trained with both supervised learning and TSSL, and transferring it to linear tasks. In this experiment, we removed the early stopping setting and trained for 100 epochs. As shown in the figure, although the training loss for supervised learning is lower than that of TSSL on the training set, supervised learning overfits too early, leading to higher validation and test losses compared to TSSL. This results in supervised training performing worse than TSSL. Therefore, TSSL effectively mitigates the overfitting phenomenon.

Fine-tuning evaluation

Tables 7, 8, 9 and 10 present the fine-tuning evaluation results for different methods across the 4 encoders. First, compared to other methods, TSSL consistently delivers the best performance across all models and datasets. Secondly, compared to the linear evaluation results, TSSL shows improved classification performance on each

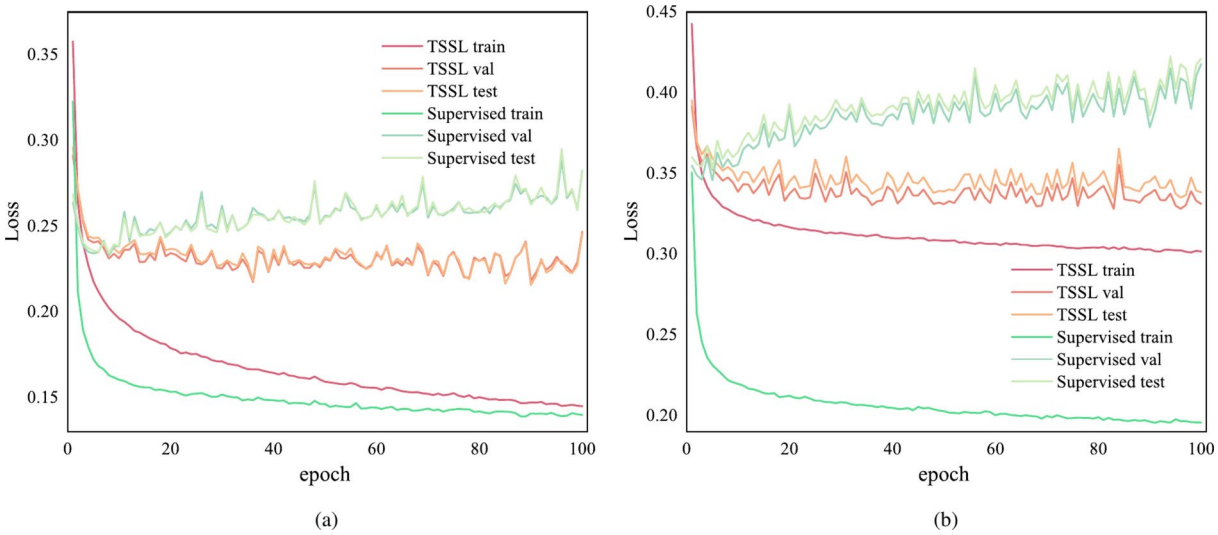


Fig. 7. Loss variation of linear evaluation classifier.

Pre-training dataset	Chapman		CPSC2018		PTB-XL	
Downstream Dataset	**CPSC2018**	**PTB-XL**	**Chapman**	**PTB-XL**	**Chapman**	**CPSC2018**
Supervised	0.878±0.003	0.863±0.014	0.971±0.004	0.866±0.006	0.965±0.006	0.859±0.036
Random Init	0.830±0.017	0.846±0.015	0.949±0.008	0.848±0.018	0.956±0.010	0.834±0.013
SimCLR	0.845±0.016	0.847±0.010	0.960±0.013	0.865±0.007	0.944±0.027	0.858±0.014
BYOL	0.823±0.006	0.816±0.015	0.908±0.012	0.831±0.007	0.878±0.014	0.807±0.012
CLOCS	0.868±0.007	0.866±0.008	0.953±0.010	0.852±0.011	0.934±0.028	0.825±0.016
MTSR	0.817±0.013	0.837±0.017	0.938±0.014	0.848±0.009	0.907±0.012	0.819±0.011
3KG	0.876±0.010	0.863±0.011	0.975±0.002	0.856±0.006	0.978±0.004	0.877±0.011
TSL	0.884±0.006	0.869±0.011	0.968±0.013	0.869±0.013	0.978±0.006	0.900±0.004
SSL	0.884±0.005	0.869±0.009	0.954±0.009	0.842±0.009	0.964±0.005	0.868±0.011
TSSL	0.885±0.002	0.868±0.013	0.970±0.011	0.876±0.007	0.977±0.011	0.893±0.007

Table 7. Fine-tuning evaluation results using the LeNet encoder.

Pre-training dataset	Chapman		CPSC2018		PTB-XL	
Downstream dataset	**CPSC2018**	**PTB-XL**	**Chapman**	**PTB-XL**	**Chapman**	**CPSC2018**
Supervised	0.883±0.016	0.881±0.06	0.977±0.003	0.885±0.005	0.966±0.018	0.888±0.008
Random Init	0.848±0.021	0.862±0.007	0.975±0.007	0.851±0.014	0.970±0.004	0.854±0.009
SimCLR	0.872±0.006	0.876±0.005	0.968±0.005	0.869±0.012	0.949±0.012	0.853±0.015
BYOL	0.806±0.017	0.833±0.008	0.880±0.027	0.831±0.015	0.859±0.004	0.823±0.012
CLOCS	0.870±0.005	0.858±0.008	0.977±0.002	0.871±0.009	0.956±0.007	0.877±0.005
MTSR	0.830±0.009	0.831±0.018	0.874±0.029	0.843±0.009	0.872±0.021	0.819±0.008
3KG	0.905±0.005	0.890±0.003	0.986±0.002	0.887±0.006	0.985±0.001	0.903±0.006
TSL	0.917±0.005	0.897±0.004	0.982±0.003	0.869±0.011	0.987±0.002	0.921±0.004
SSL	0.917±0.004	0.897±0.003	0.977±0.004	0.856±0.015	0.973±0.001	0.865±0.012
TSSL	0.913±0.005	0.896±0.008	0.986±0.001	0.897±0.004	0.986±0.001	0.911±0.007

Table 8. Fine-tuning evaluation results using the AlexNet encoder.

Pre-training dataset	Chapman		CPSC2018		PTB-XL	
Downstream dataset	CPSC2018	PTB-XL	Chapman	PTB-XL	Chapman	CPSC2018
Supervised	0.872±0.011	0.861±0.014	0.992±0.001	0.877±0.011	0.966±0.005	0.888±0.014
Random Init	0.839±0.014	0.862±0.010	0.988±0.003	0.857±0.013	0.985±0.008	0.832±0.037
SimCLR	0.898±0.014	0.895±0.006	0.989±0.003	0.884±0.012	0.979±0.012	0.890±0.010
BYOL	0.803±0.036	0.841±0.006	0.953±0.008	0.847±0.011	0.905±0.043	0.795±0.043
CLOCS	0.866±0.010	0.847±0.010	0.974±0.009	0.872±0.006	0.979±0.004	0.881±0.006
MTSR	0.782±0.056	0.815±0.015	0.929±0.019	0.815±0.024	0.919±0.028	0.781±0.034
3KG	0.899±0.014	0.871±0.014	0.995±0.001	0.888±0.010	0.995±0.001	0.891±0.027
TSL	0.913±0.006	0.893±0.006	0.995±0.000	0.889±0.006	0.994±0.001	0.924±0.005
SSL	0.918±0.004	0.892±0.008	0.986±0.006	0.883±0.006	0.990±0.001	0.870±0.012
TSSL	0.909±0.010	0.879±0.008	0.994±0.001	0.887±0.003	0.994±0.001	0.911±0.009

Table 9. Fine-tuning evaluation results using the VGG encoder.

Pre-training dataset	Chapman		CPSC2018		PTB-XL	
Downstream dataset	CPSC2018	PTB-XL	Chapman	PTB-XL	Chapman	CPSC2018
Supervised	0.870±0.012	0.872±0.007	0.985±0.002	0.869±0.012	0.982±0.004	0.881±0.007
Random Init	0.802±0.026	0.846±0.010	0.960±0.023	0.851±0.014	0.983±0.004	0.842±0.015
SimCLR	0.865±0.012	0.867±0.008	0.960±0.016	0.866±0.016	0.951±0.014	0.840±0.034
BYOL	0.725±0.021	0.783±0.027	0.829±0.025	0.807±0.036	0.810±0.016	0.738±0.015
CLOCS	0.867±0.006	0.856±0.011	0.976±0.008	0.858±0.005	0.975±0.002	0.878±0.005
MTSR	0.693±0.026	0.759±0.023	0.791±0.017	0.749±0.030	0.771±0.016	0.701±0.021
3KG	0.904±0.011	0.893±0.007	0.993±0.001	0.885±0.009	0.994±0.001	0.905±0.014
TSL	0.909±0.016	0.896±0.007	0.994±0.001	0.892±0.005	0.994±0.000	0.920±0.003
SSL	0.844±0.027	0.860±0.010	0.985±0.005	0.855±0.016	0.977±0.008	0.866±0.019
TSSL	0.909±0.004	0.874±0.013	0.993±0.001	0.888±0.009	0.993±0.000	0.907±0.003

Table 10. Fine-tuning evaluation results using the ResNet encoder.

dataset after fine-tuning. The best performance on the three datasets (Chapman, CPSC2018, PTB-XL) improved from 0.984, 0.882, and 0.877 to 0.995, 0.921, and 0.897, respectively, indicating that the extracted representations have strong robustness and transferability.

Ablation analysis

As seen in Tables 3, 4, 5 and 6, the representations extracted using only spatial self-supervision (SSL) outperform those from a randomly initialized encoder, indicating that SSL has a certain effect. However, its performance on CPSC and PTB-XL is relatively poor, with suboptimal representation extraction. This is because spatial self-supervision only maintains the correlation between leads, lacking effective measures to extract the latent, efficient features of ECGs. TSL performs well across various models and datasets, indicating that TSL is an effective self-supervised learning method for ECG classification. By combining TSL and SSL, TSSL achieves better results than either method alone, validating our approach of using spatial self-supervision to maintain lead correlation during temporal self-supervision. Figures 8, 9 and 10 show the absolute differences in cosine similarity between leads for a particular signal representation extracted by TSL and TSSL, as well as the classification probability, when using the VGG encoder for linear evaluation on the three datasets. This demonstrates that maintaining inter-lead correlation when extracting self-supervised signal representations helps improve classification accuracy.

As shown in Tables 7, 8, 9 and 10, in fine-tuning scenarios, representations extracted by temporal self-supervised learning (TSL) outperform those extracted by temporal-spatial self-supervised learning (TSSL). This is because too many supervision conditions can reduce the model's ability to generalize across data distributions, weakening the transferability of the representations. In practical applications, TSL and TSSL can be used flexibly depending on the specific scenario.

Analysis of dependency on labels

The purpose of proposing TSSL is to reduce the dependency of deep learning methods on labels in the field of automatic ECG signal detection, where acquiring labels is challenging. In this section, we use VGG as the encoder, training the classifier with only a portion of the labels. We compare the classification performance of supervised learning and TSSL under different label quantities and examine how label quantity affects classification performance.

As seen in Fig. 11, TSSL exhibits more robust performance compared to supervised learning methods when the number of labels varies. Particularly on the Chapman dataset, which has a lower classification difficulty, TSSL

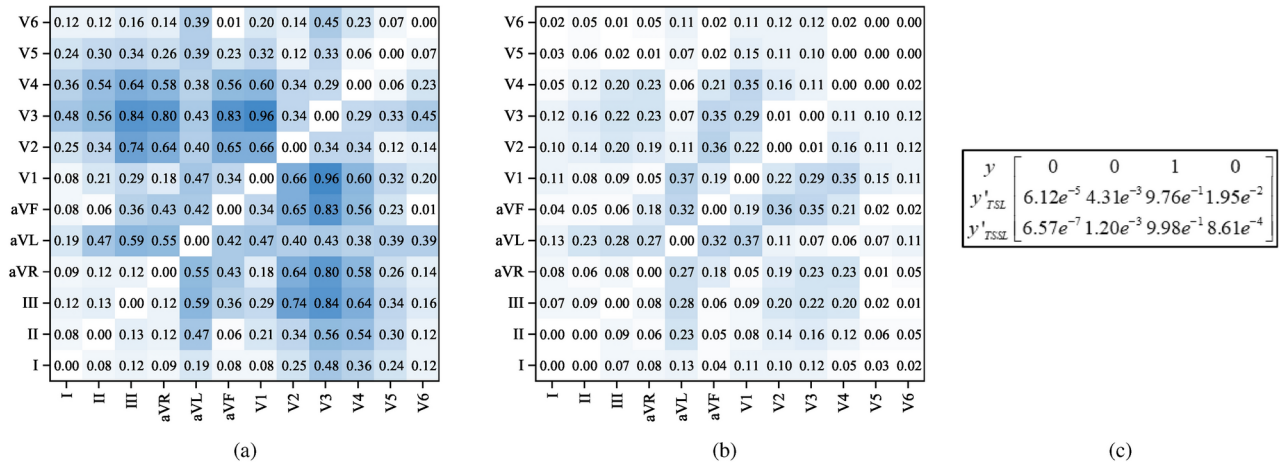


Fig. 8. Absolute error of cosine similarity between signals and representations across leads on Chapman. Train the encoder using different TSL and TSSL respectively, obtain the absolute differences between the correlations of the representations across leads and those of the original signals, as well as the classification probabilities on different classes after training the representations.

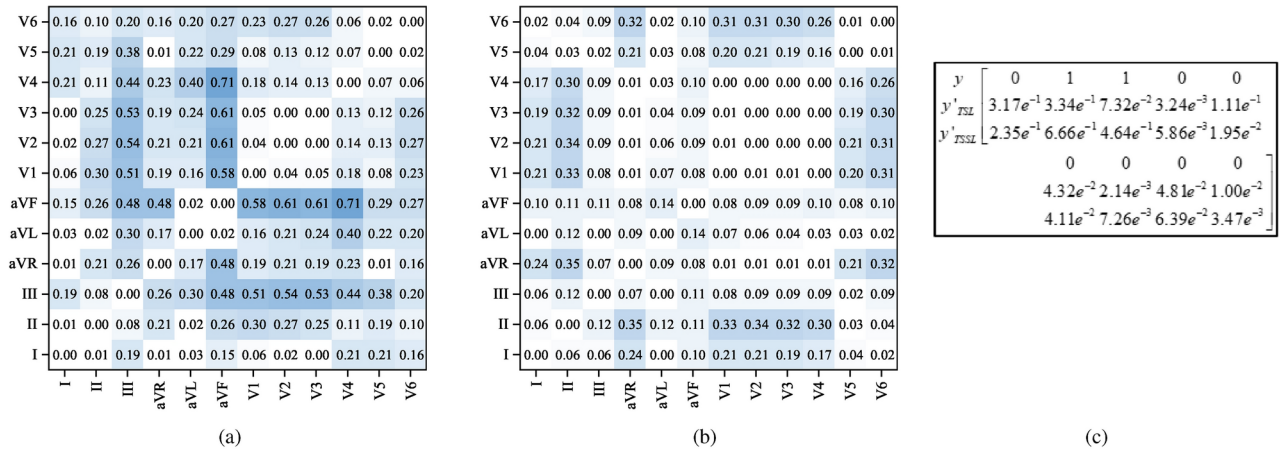


Fig. 9. Absolute error of cosine similarity between signals and representations across leads on CPSC2018.

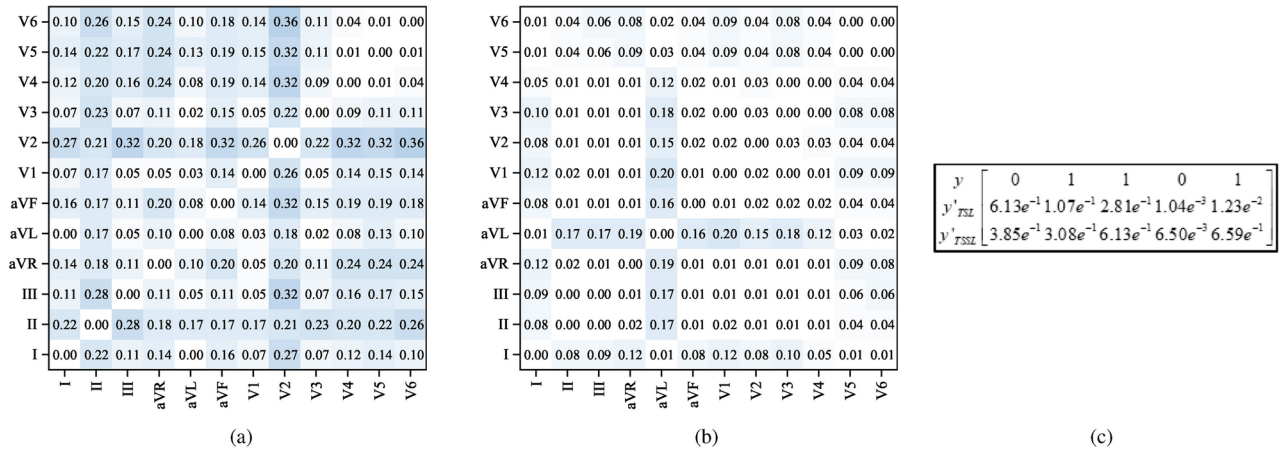


Fig. 10. Absolute error of cosine similarity between signals and representations across leads on PTB-XL.

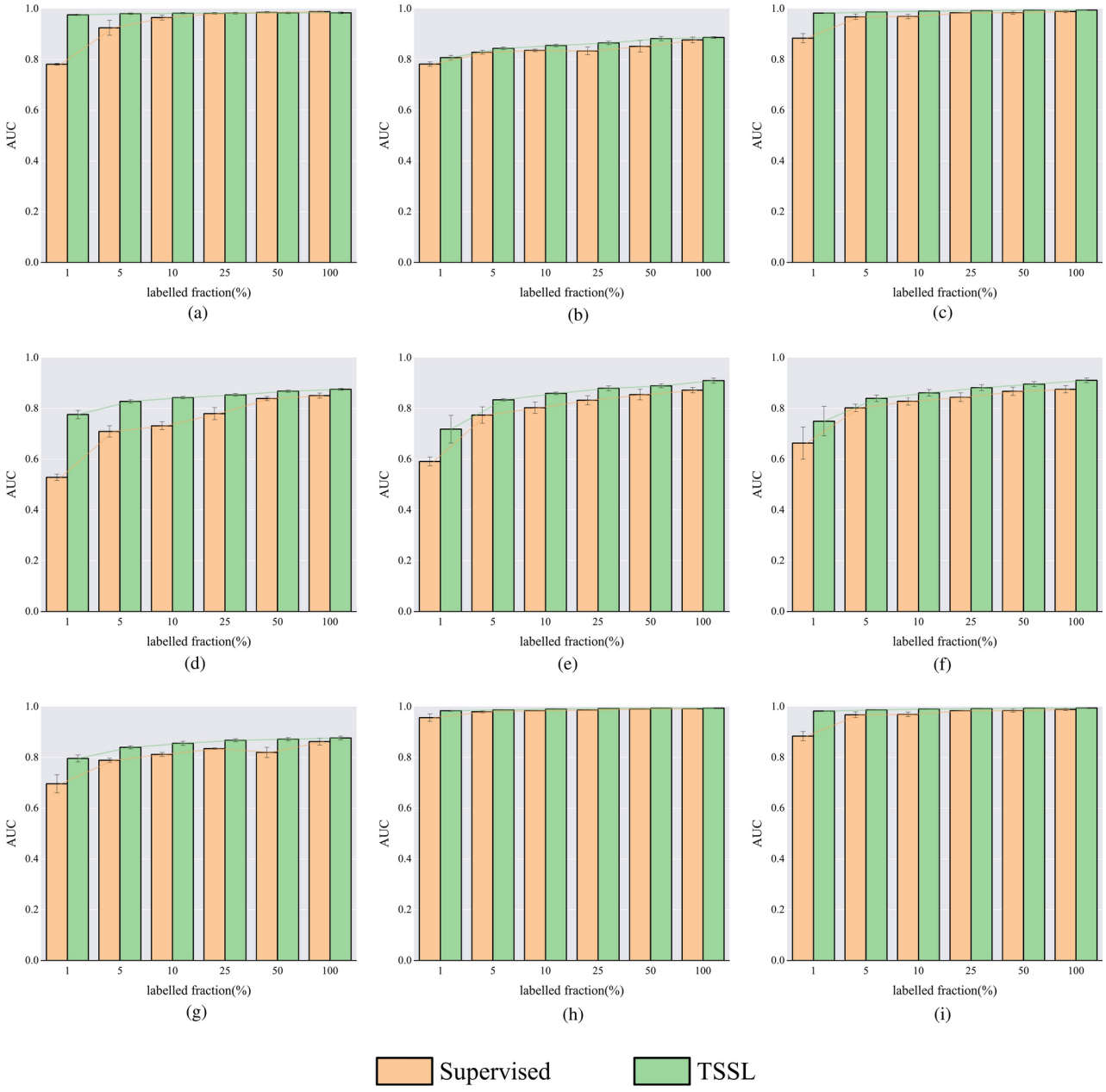


Fig. 11. Effect of label quantity on classification performance, classification results trained with different available labels.

requires only 1% of the labels to achieve performance close to that of supervised training with 100% of the labels, whether in linear or classification tasks. On the more complex CPSC2018 and PTB-XL datasets, this proportion is 10%. In other words, compared to supervised training, TSSL can achieve nearly equivalent performance with only 10% of the labeling effort, thus reducing dependency on label quantity.

Discussion

This study introduces a temporal-spatial self-supervised learning methods (TSSL) that exhibits significant advantages in automatic electrocardiogram (ECG) detection. By integrating temporal self-supervised learning (TSL) and spatial self-supervised learning (SSL), TSSL effectively harnesses the temporal and spatial characteristics of ECG signals, thereby enhancing the quality and robustness of feature representations. Notably, TSSL achieves a remarkable performance improvement of over 15% on the complex PTB-XL dataset compared to other methods, demonstrating its superior adaptability in handling intricate ECG patterns. Additionally, TSSL consistently delivers excellent performance across various encoder architectures, including both the simple LeNet and the more complex ResNet, underscoring its high robustness and broad applicability.

Ablation analysis further validates the effectiveness of the TSSL strategy. Compared to using TSL or SSL individually, TSSL consistently outperforms them in all experiments, highlighting the crucial synergy between

temporal and spatial self-supervision in enhancing ECG representations. This finding not only underscores the innovative nature of the TSSL method but also offers new insights for the future application of self-supervised learning in ECG analysis.

Furthermore, on the CPSC2018 and PTB-XL datasets, TSSL surpasses supervised training methods, showcasing its potential and practical value in real-world applications. This superior performance is attributed to TSSL's ability to effectively mitigate overfitting during training, thereby improving the model's generalization capability.

As a self-supervised learning strategy, TSSL is applicable to any ECG signal classification model and allows for a balanced design of encoders based on computational resources and performance requirements in practical applications. This flexibility highlights its tremendous potential in real-world ECG diagnostic settings. TSSL not only outperforms existing methods in terms of performance but also offers multiple innovative advantages in method design and practical application, providing an efficient, robust, and practical solution for automatic ECG detection.

A limitation of this study is that it only examines fixed 10-second input data and has not yet validated its performance on long-term ECG recordings, which may hinder its deployment in clinically relevant tasks. Future work will focus on training and testing the model with long-term ECG data to better address clinical needs.

Moreover, TSSL can be extended to other multi-lead biological signals (such as electroencephalogram signals) and multimodal biological signals collected simultaneously (such as ECG, EEG, or EMG signals). Whether dealing with multi-lead or multimodal biological signals, each individual signal should contain personal identity information over time, ensuring consistency, while different simultaneously collected signals exhibit certain correlations. Although the term "spatial" may not be entirely appropriate in this context, it reflects characteristics similar to the spatial correlations discussed in this paper. Therefore, future research can expand the TSSL method to other domains of biological signals.

Conclusion

In this paper, we proposed a novel self-supervised learning model based on the temporal and spatial properties of ECG signals. We designed temporal and spatial self-supervised learning schemes to learn rich and generalizable representations from unlabeled multi-lead ECG signals, aiming to reduce the dependency of deep learning on data labels and improve arrhythmia diagnosis. Through experiments on multiple datasets and encoders, we validated the robustness and transferability of TSSL. Compared to traditional supervised learning methods, TSSL exhibits more robust performance when the number of labels varies, particularly on complex datasets, where it achieves results close to supervised training with only 10% of labeled data. Additionally, TSSL combines the strengths of temporal self-supervision (TSL) and spatial self-supervision (SSL), maintaining inter-lead correlations during representation extraction while effectively reducing overfitting. These results indicate that TSSL is an efficient and practical self-supervised learning method, capable of providing reliable representation extraction in ECG signal classification while significantly reducing the need for large-scale labeled data.

Data availability

The ECGs used in this study were from open source databases and available in https://physionet.org/content/ptb-xl/1.0.1/ (PTB-XL); https://figshare.com/collections/ChapmanECG/4560497/2 (Chapman) and https://physionet.org/content/challenge-2020/1.0.2/training/cpsc_2018/ (CPSC2018).

Received: 30 September 2024; Accepted: 10 February 2025
Published online: 19 February 2025

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Author contributions

W.C.: Carried designed and implemented the computer programs to a analyze data and obtain the results, Wrote the first draft manuscript. H.W.: Provide guidance on paper writing.L.Z.: Provide guidance for the experiment. M.Z.: Provide guidance for the experiment. All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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